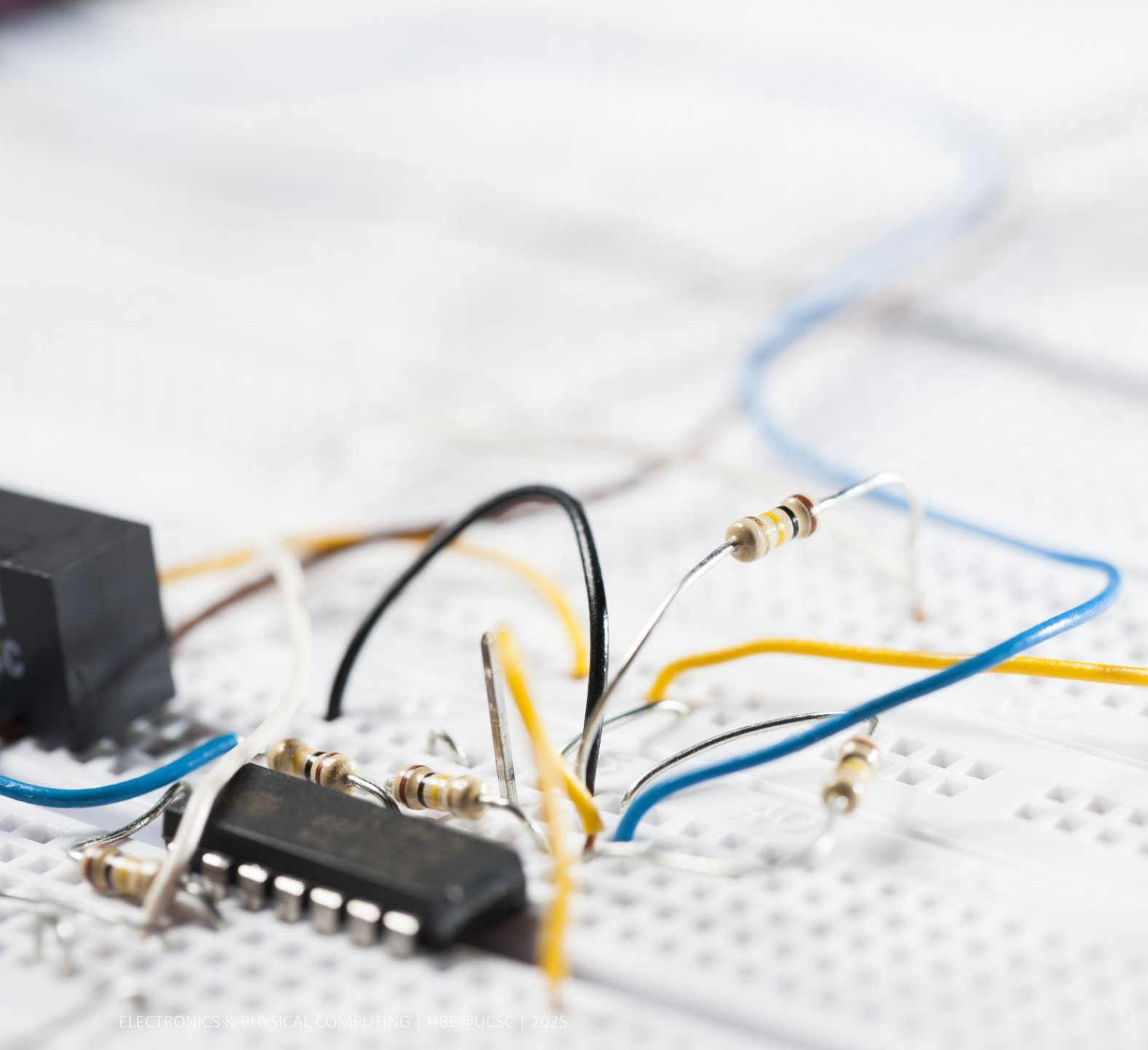


SCS2315 – Electronics & Physical Computing

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Lesson 1-1: Foundations of Electronics and Signals

Electronics & Physical Computing



What is Electronics?

Electronics is the branch of science and technology that deals with the study, design, and use of devices and systems that operate by controlling the flow of **electrons** or other charge **carriers** (like holes in semiconductors).

In simpler terms:
Electronics is about using **electric current** to process, transmit, or store information.

Key Concepts in Electronics

Electric Current – The flow of electric charge, usually through conductors like wires.

Circuits – Paths through which electric current flows. Circuits can be simple (like a battery and a bulb) or complex (like inside a smartphone).

Components – Basic building blocks like:

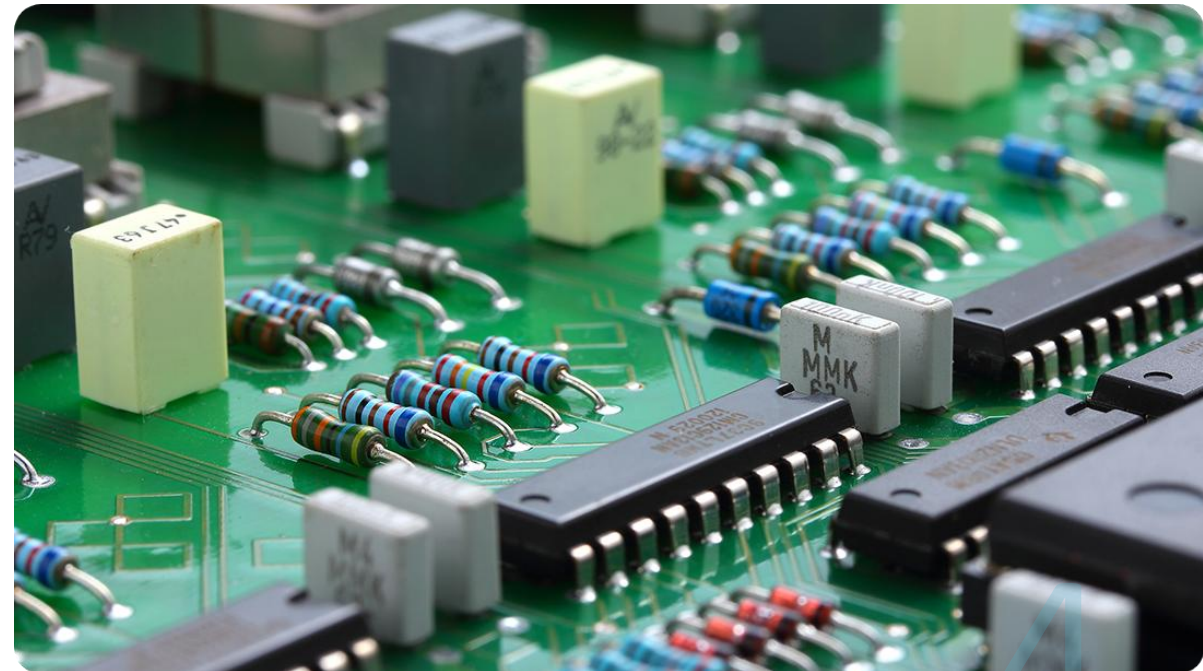
- Resistors (control current)

- Capacitors (store energy)

- Inductors (store energy in magnetic fields)

- Diodes (let current flow in one direction)

- Transistors (act as switches or amplifiers)

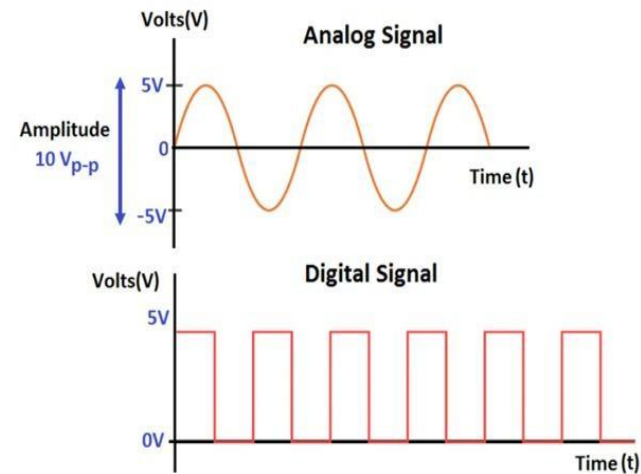


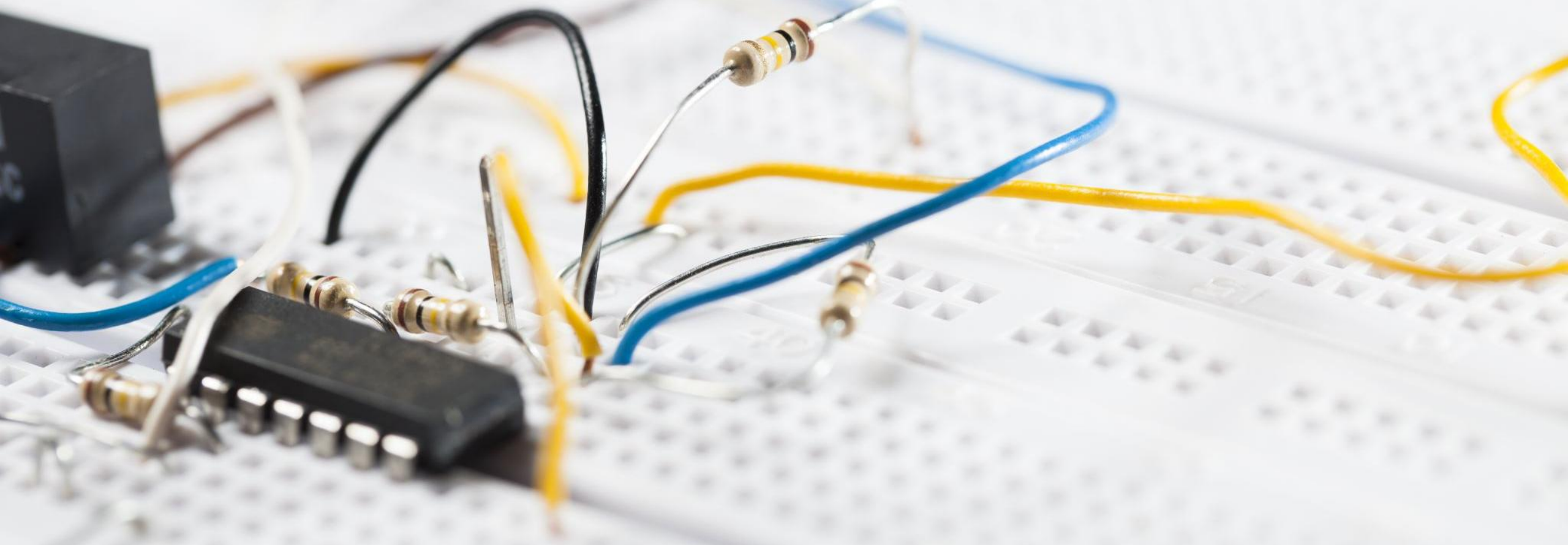
Key Concepts in Electronics

Analog vs. Digital Electronics:

Analog deals with continuous signals (e.g., audio signals).

Digital deals with discrete signals (e.g., 0s and 1s in computers).

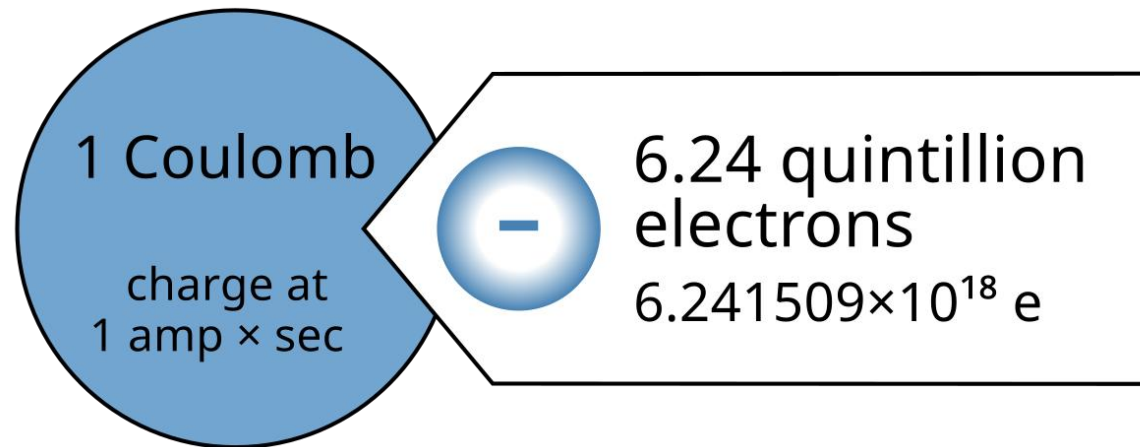




Electrical Quantities

1. Charge (Q)

- **Definition:** The fundamental property of matter responsible for electric phenomena.
- **Unit: Coulomb (C)**
- **Example:** A single electron has a charge of about -1.6×10^{-19} C.



2. Current (I)

- **Definition:** The rate of flow of electric charge.

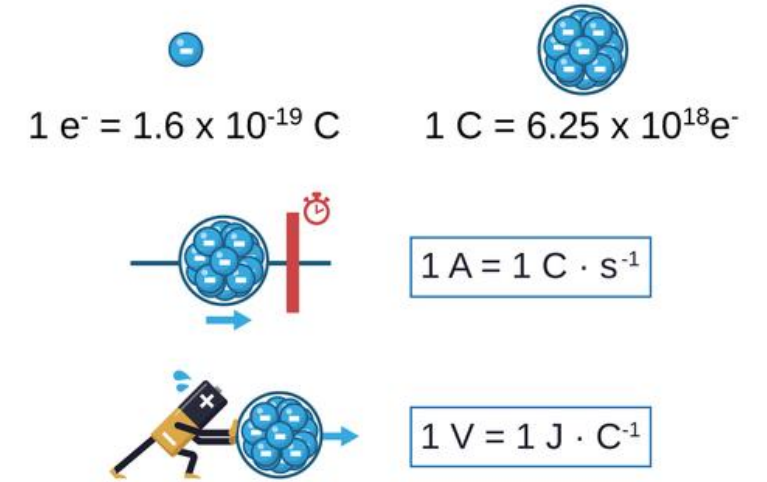
- **Formula:**

$$I = \frac{Q}{t}$$

where Q is charge, t is time.

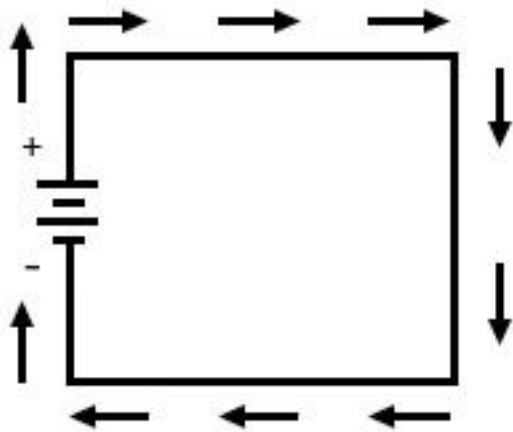
- **Unit:** Ampere (A)

- **Example:** If 1 coulomb of charge flows in 1 second, that's 1 ampere of current.



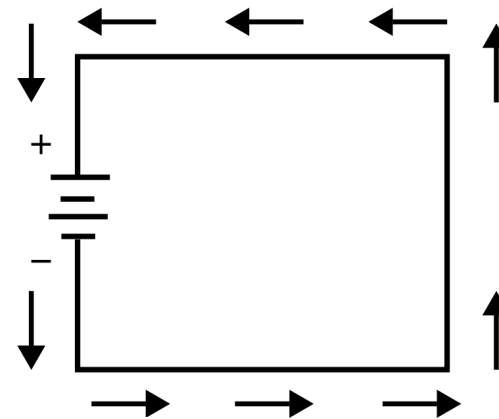
Note

Conventional Flow Notation

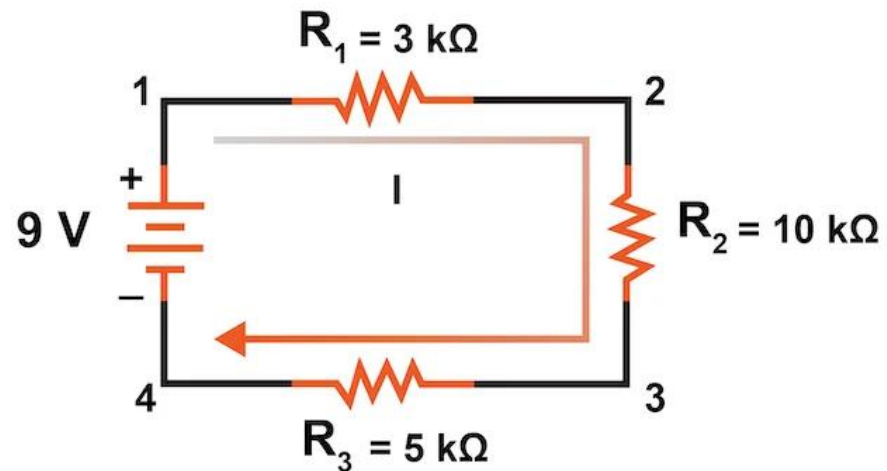


Electric charge moves from the positive (surplus) side of the battery to the negative (deficiency) side.

Electron Flow Notation



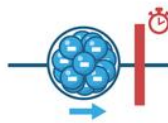
Electron charge moves from the negative (surplus) side of the battery to the positive (deficiency) side.



3. Voltage (V)

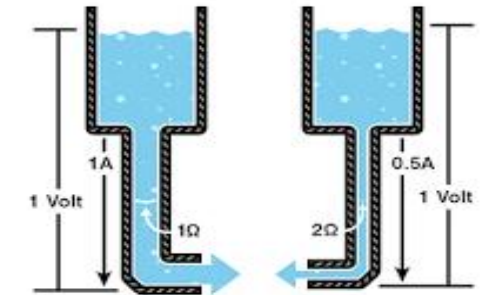
 $1 e^- = 1.6 \times 10^{-19} C$

 $1 C = 6.25 \times 10^{18} e^-$

 $1 A = 1 C \cdot s^{-1}$

 $1 V = 1 J \cdot C^{-1}$

- **Definition:** The **potential difference** between two points; it's the “push” that drives current.
- **Formula:**
$$V = \frac{W}{Q}$$
where W is work or energy in joules, Q is charge.
- **Unit: Volt (V)**
- **Example:** A 9V battery can push charges through a circuit with 9 volts of potential difference.



4. Resistance (R)

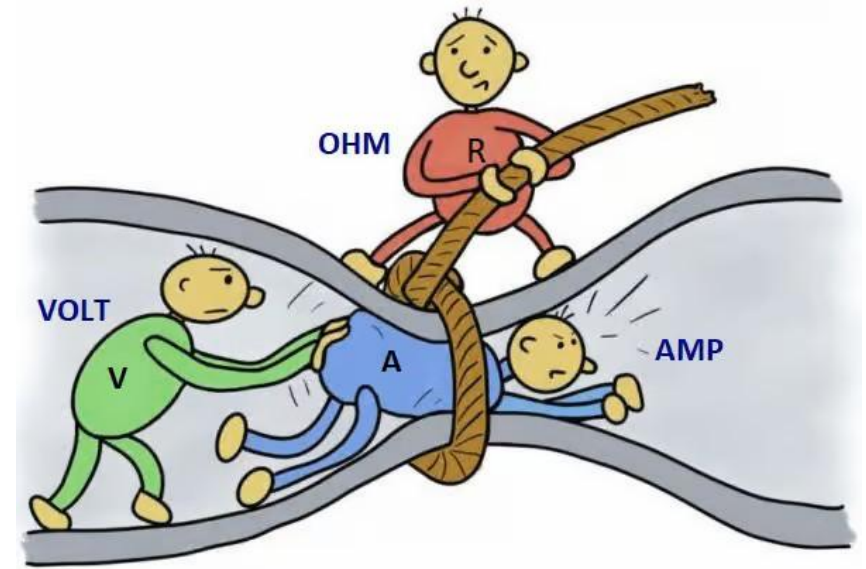
- **Definition:** The **opposition to the flow of current** in a material.

- **Formula:**

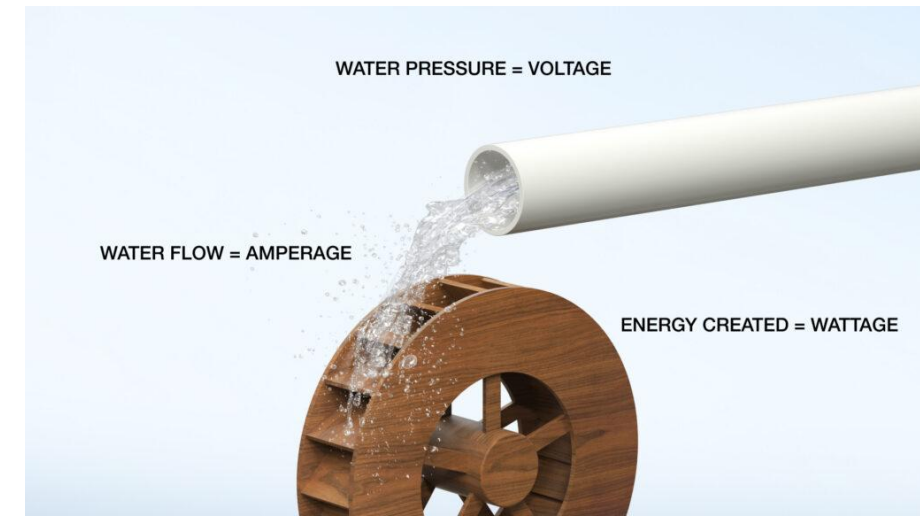
$$R = \frac{V}{I}$$

- **Unit:** Ohm (Ω)

- **Example:** A resistor that allows 1 amp of current to flow when 5 volts are applied has a resistance of 5 ohms.



5. Power (P)



- **Definition:** The rate at which electrical energy is consumed or generated.
- **Formula:**
$$P = VI$$

(Also: $P = I^2 R$ or $P = \frac{V^2}{R}$)
- **Unit:** Watt (W)
- **Example:** A 60W bulb consumes 60 joules of energy every second.

6. Energy (E)

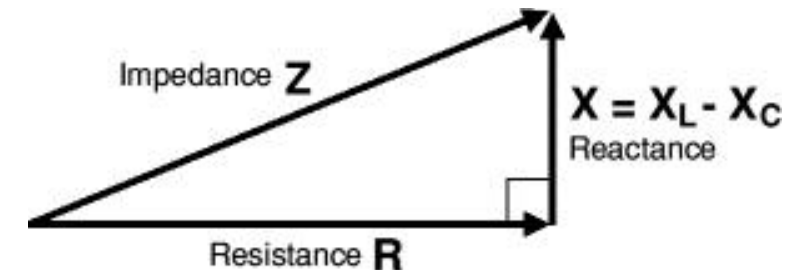
- **Definition:** The **capacity to do work**; in electronics, it's how much power is used over time.
- **Formula:**
$$E = Pt$$
- **Unit:** Joule (J) or Watt-hour (Wh)
- **Example:** A 100W appliance running for 2 hours uses 200 Wh of energy.



7. Reactance (X)

- **Definition:** The opposition to AC current caused by inductors and capacitors (not resistors).
- **Types:**
 - Inductive Reactance (X_L): $X_L = 2\pi fL$
 - Capacitive Reactance (X_C): $X_C = \frac{1}{2\pi fC}$
- **Unit:** Ohm (Ω)
- **Note:** Reactance depends on frequency (f) of the signal.

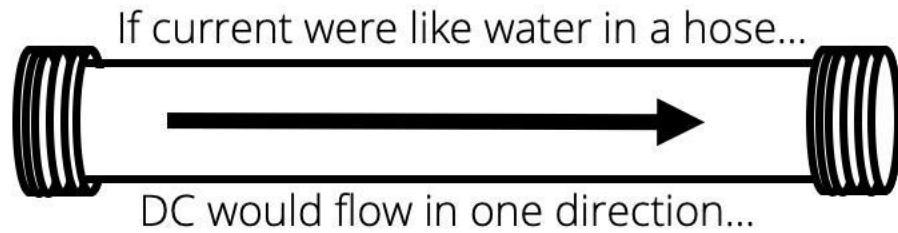
Resistor  R = 100 Ω X = 0 Ω Z = 100 $\Omega \angle 0^\circ$	Inductor  R = 0 Ω X = 100 Ω Z = 100 $\Omega \angle 90^\circ$	Capacitor  R = 0 Ω X = 100 Ω Z = 100 $\Omega \angle -90^\circ$
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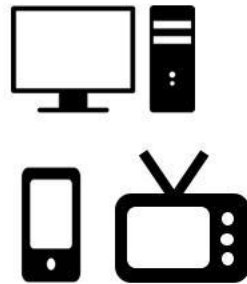
$$\text{Impedance, } Z = \sqrt{R^2 + X^2}$$

Alternating Current vs Direct Current

DC

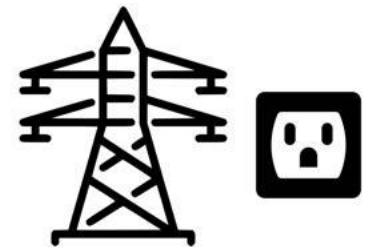
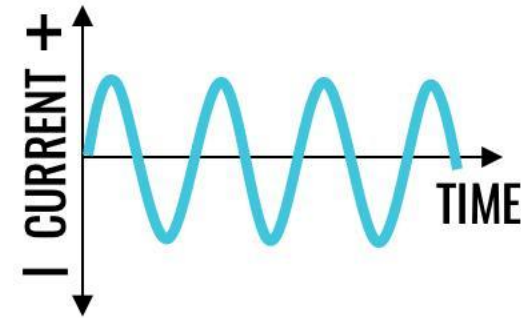
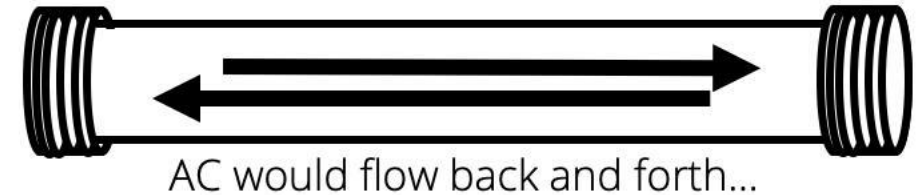


FREEING
ENERGY

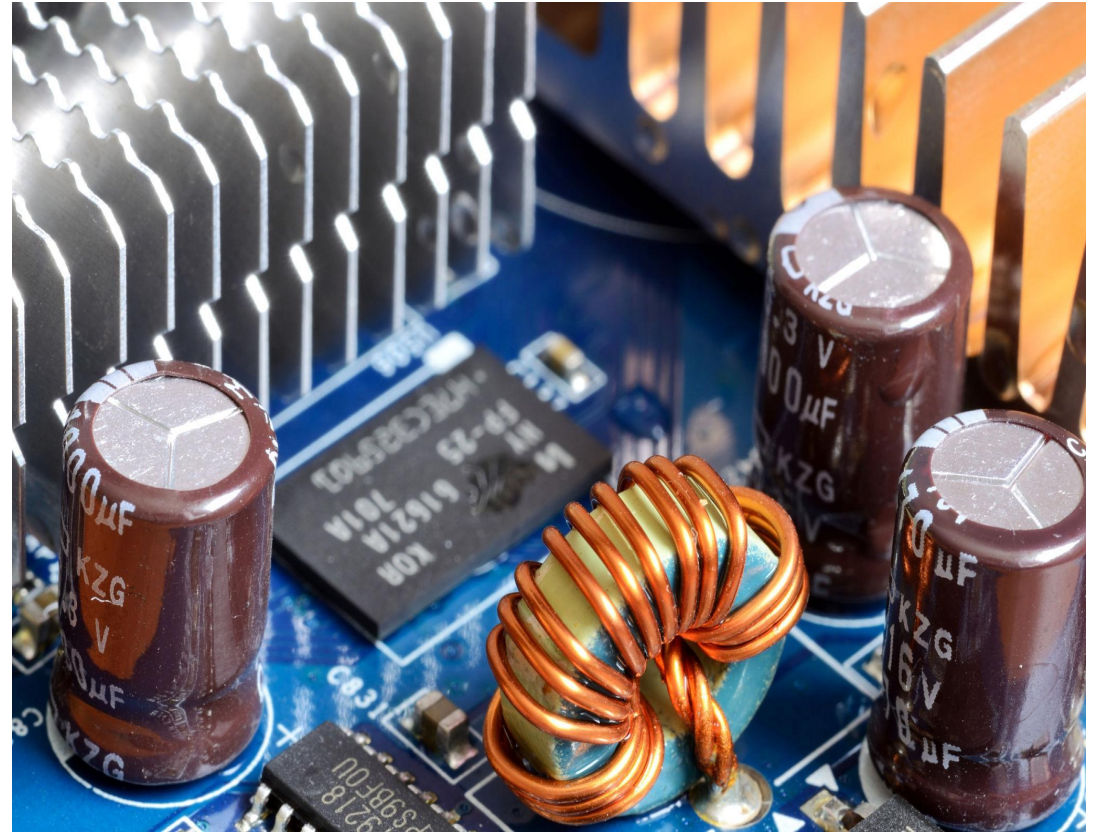


Things that use DC

AC



Things that use AC



8. Impedance (Z)

- **Definition:** The total opposition to AC that includes both resistance and reactance.

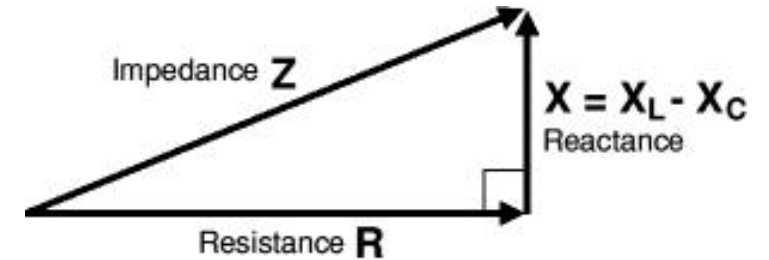
- **Formula:**

$$Z = R + jX$$

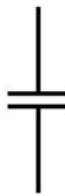
(where j is the imaginary unit, indicating phase difference)

- **Unit:** Ohm (Ω)

- **Example:** In AC circuits, impedance determines how signals behave through components.



$$\text{Impedance, } Z = \sqrt{R^2 + X^2}$$

Resistor 100 Ω	Inductor 100 mH 159.15 Hz	Capacitor 10 μF 159.15 Hz
		
$R = 100 \Omega$ $X = 0 \Omega$ $Z = 100 \Omega \angle 0^\circ$	$R = 0 \Omega$ $X = 100 \Omega$ $Z = 100 \Omega \angle 90^\circ$	$R = 0 \Omega$ $X = 100 \Omega$ $Z = 100 \Omega \angle -90^\circ$

Electrical Quantity	Water Analogy	Meaning / Visualization
Charge (Q)	Amount of water	Total "stuff" flowing
Current (I)	Flow rate of water (liters/sec)	How fast the water flows
Voltage (V)	Water pressure	The "push" behind the flow
Resistance (R)	Narrow pipe	Harder to flow through
Power (P)	Water force $\times$ flow rate	Work being done (e.g., spinning a waterwheel)
Energy (E)	Total water delivered over time	How much total work is done
Reactance (X)	Flexible or elastic pipe	Temporarily resists changes in flow (frequency dependent)
Impedance (Z)	Combined narrow and elastic pipe	Total opposition to flow in AC circuits

Electrical Water Analogy

Exercise

Q1. A current of **2 A** flows through a conductor for **3 minutes**. Find the **total charge** that passes through the conductor.

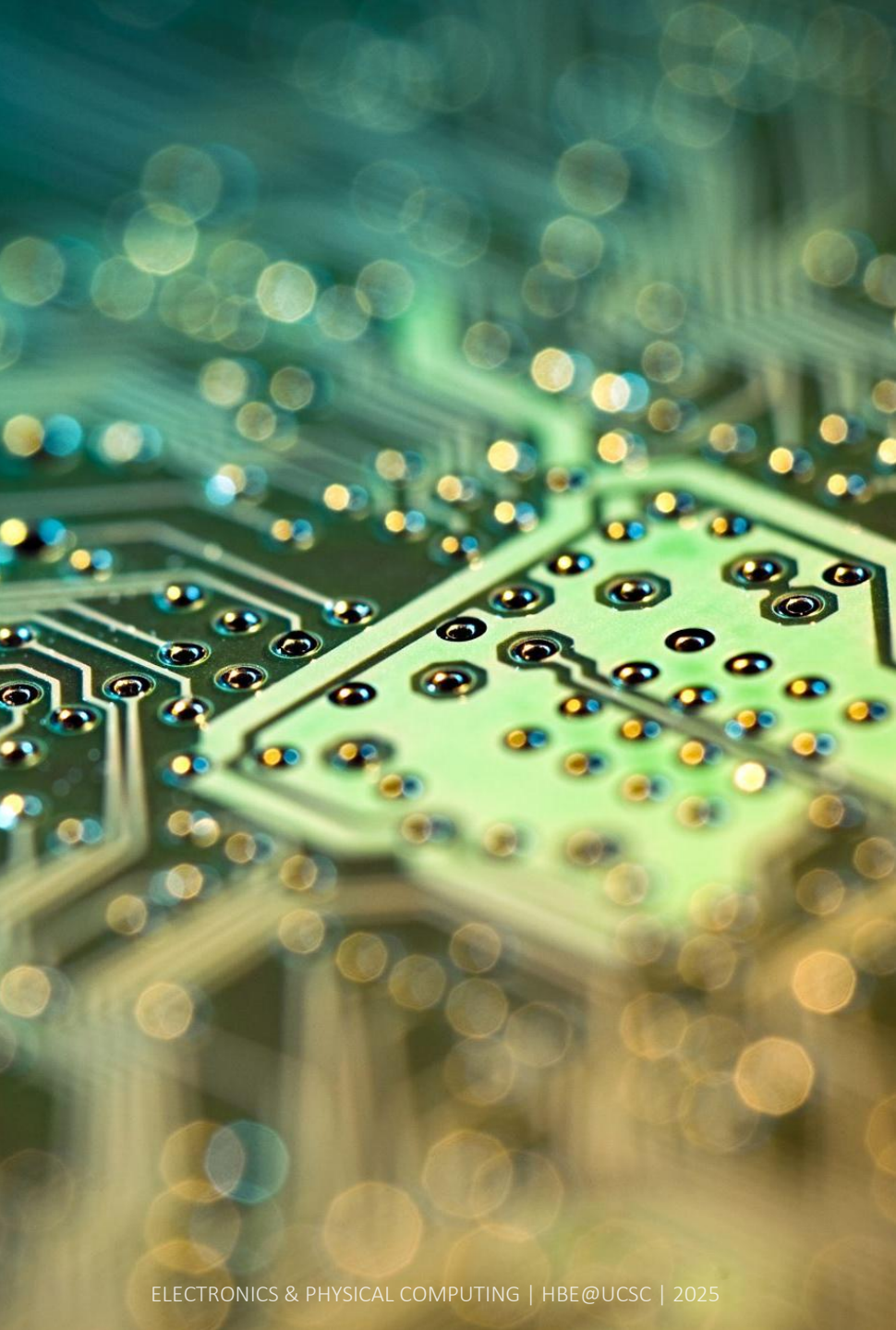
Q2. How many **electrons** flow through a wire when a **charge of 0.5 C** passes? (Charge of one electron = $1.6 \times 10^{-19} \text{ C}$)

$$Q = I \times t$$

$$Q = 2 \text{ A} \times (3 \times 60) \text{ s} = 2 \times 180 = 360 \text{ C}$$

$$n = \frac{Q}{e}$$

$$n = \frac{0.5}{1.6 \times 10^{-19}} = 3.125 \times 10^{18}$$



Basic Circuit Laws

Ohm's Law

$$V = I \times R$$

Ohm's Law defines the relationship between **voltage (V)**, **current (I)**, and **resistance (R)** in an electrical circuit.

V (Voltage): Potential difference across the resistor (volts, V)

I (Current): Flow of electric charge (amperes, A)

R (Resistance): Opposition to current flow (ohms, Ω)

Note: The symbols “E” and “V” are interchangeable for the most part, although some texts reserve “E” (stands for “Electromotive force”) to represent voltage across a source (such as a battery or generator) and “V” to represent voltage across anything else.

Exercise 1 (Ohm's Law)

A 18 V battery is connected across a resistor of 6 Ω .

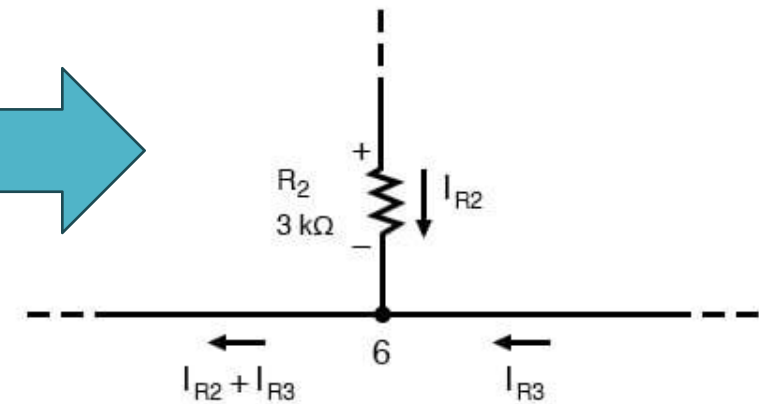
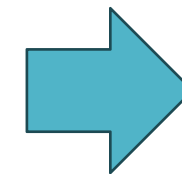
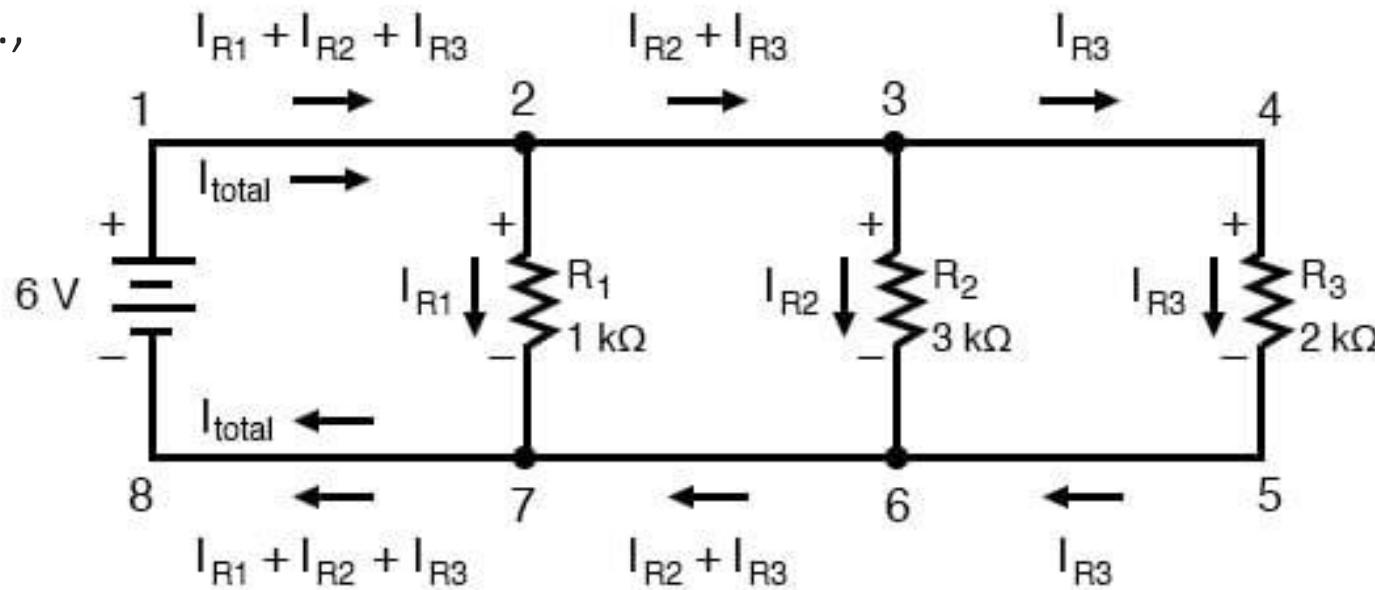
1. Find the current flowing through the resistor.
2. Calculate the power dissipated by the resistor.
3. If the resistor is connected for 4 minutes, what happens due to energy dissipation?

Kirchhoff's Current Law (KCL)

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

At any junction (node) in a circuit, the **sum of currents entering the node equals the sum of currents leaving the node**, or in other words, the **algebraic sum of all currents entering and exiting a node must equal zero**.

E.g.,



Exercise 1 (KCL)

Problem: A node has a current source injecting 6.0 A into it. The node is connected to ground through $R_1 = 2.0 \text{ }\Omega$ and $R_2 = 3.0 \text{ }\Omega$. Find the node voltage V .

KCL: currents leaving node through resistors = injected current

$$\frac{V}{R_1} + \frac{V}{R_2} = 6.0$$

Answer: $V = 7.2 \text{ V}$.

Exercise 2 (KCL)

Problem: Two nodes A and B (relative to ground) are connected as follows:

- Node A to ground through $R_A = 2 \Omega$.
- Node B to ground through $R_B = 6 \Omega$.
- Node A connected to Node B by $R_{AB} = 4 \Omega$.
- Current source $I_1 = 3.0 \text{ A}$ injects into Node A.
- Current source $I_2 = 1.0 \text{ A}$ injects into Node B.

Find V_A and V_B .

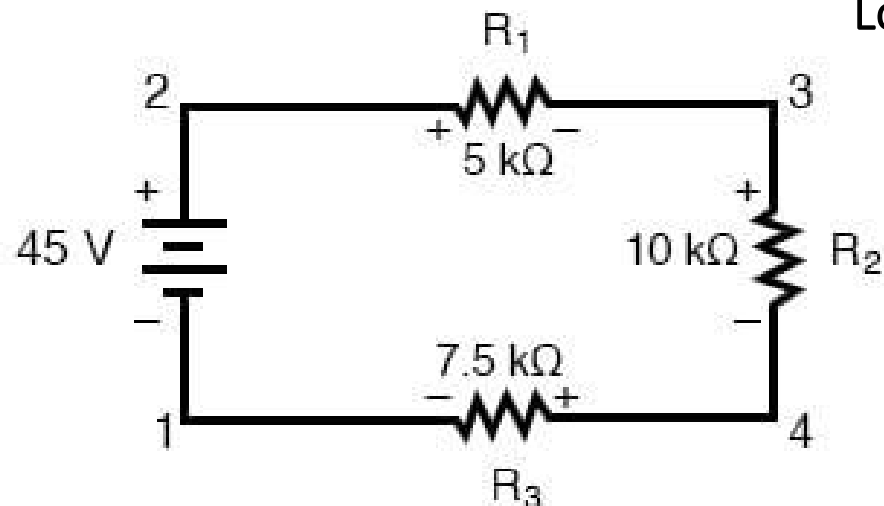
$$3 = \frac{V_A}{2} + \frac{V_A - V_B}{4}, \quad 1 = \frac{V_B}{6} + \frac{V_B - V_A}{4}. \quad \text{Answer: } V_A = 6.0 \text{ V}, \quad V_B = 6.0 \text{ V}.$$

$$\sum V = 0$$

Kirchhoff's Voltage Law (KVL)

In any closed loop in a circuit, the **sum of all voltages** (rises and drops) equals zero, or in other words, the **algebraic sum of all voltages** in a loop must equal zero.

E.g.,



Loop Order: 2-1-4-3-2

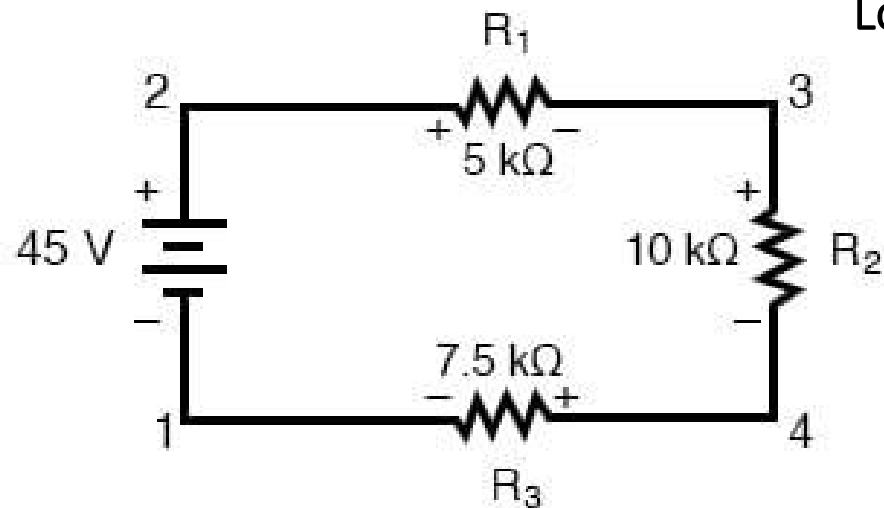
$E_{2-1} = +45 \text{ V}$	<i>voltage from point 2 to point 1</i>
$E_{3-2} = -10 \text{ V}$	<i>voltage from point 3 to point 2</i>
$E_{4-3} = -20 \text{ V}$	<i>voltage from point 4 to point 3</i>
$+ E_{1-4} = -15 \text{ V}$	<i>voltage from point 1 to point 4</i>
0 V	

$$\sum V = 0$$

Kirchhoff's Voltage Law (KVL)

In any closed loop in a circuit, the **sum of all voltages** (rises and drops) equals zero, or in other words, the **algebraic sum of all voltages** in a loop must equal zero.

E.g.,



Loop Order: 2-3-4-1-2

$E_{2-3} = +10 \text{ V}$	<i>voltage from point 2 to point 3</i>
$E_{1-2} = -45 \text{ V}$	<i>voltage from point 1 to point 2</i>
$E_{4-1} = +15 \text{ V}$	<i>voltage from point 4 to point 1</i>
$+ E_{3-4} = +20 \text{ V}$	<i>voltage from point 3 to point 4</i>
0 V	

Exercise 1 (KVL)

Problem: A single loop contains a 12 V battery and two series resistors $R_1 = 2 \Omega$ and $R_2 = 4 \Omega$. Find the loop current I (clockwise).

KVL (following the loop in the direction of current):

$$+12 - I \cdot 2 - I \cdot 4 = 0$$

Answer: $I = 2.000 \text{ A}$ (clockwise).

Exercise 2 (KVL)

A circuit has two resistors connected in series with a 24 V battery.

- $R_1 = 4\Omega$
- $R_2 = 8\Omega$

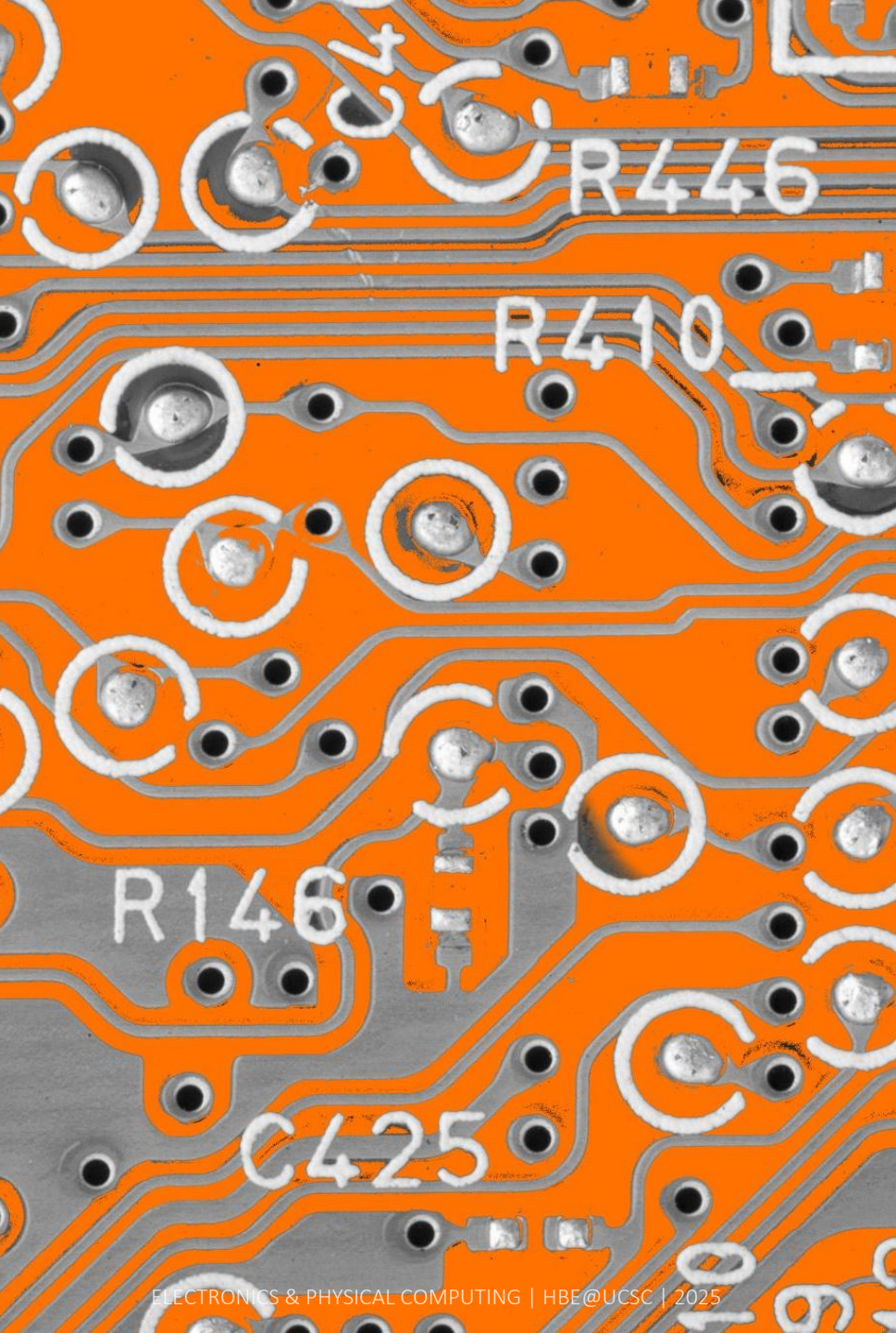
Find:

1. The current in the circuit
2. Voltage across each resistor
3. Power dissipated by each resistor

Exercise 3 (Theories Combined)

A 12.0 V battery is connected through a series resistor $R_1 = 4.0\ \Omega$ to node A. From node A to ground there are two resistors in parallel: $R_2 = 6.0\ \Omega$ and $R_3 = 12.0\ \Omega$. In addition, a current source of $I_s = 2.0\text{ A}$ injects current into node A (from ground up into node A). Find node voltage V_A , the branch currents I_{R2} , I_{R3} (to ground), and the series current I_{series} coming from the battery into node A.

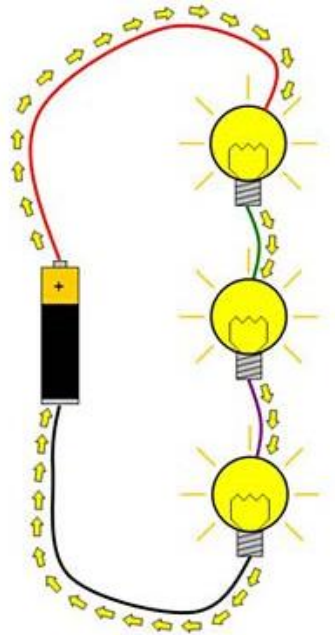
- Node voltage: $V_A = 10.0\text{ V}$.
- Currents to ground: $I_{R2} = \frac{5}{3}\text{ A} \approx 1.6667\text{ A}$, $I_{R3} = \frac{5}{6}\text{ A} \approx 0.8333\text{ A}$.
- Series current from battery: $I_{series} = 0.50\text{ A}$.



Circuit Types

Series Circuits

Components are connected **end-to-end**, so the **same current flows** through all of them.



🔑 Key Characteristics:

- **Current:** Same through all components
 $I_{\text{total}} = I_1 = I_2 = \dots$
- **Voltage:** Splits across components
 $V_{\text{total}} = V_1 + V_2 + \dots$
- **Resistance:** Adds up
 $R_{\text{total}} = R_1 + R_2 + \dots$

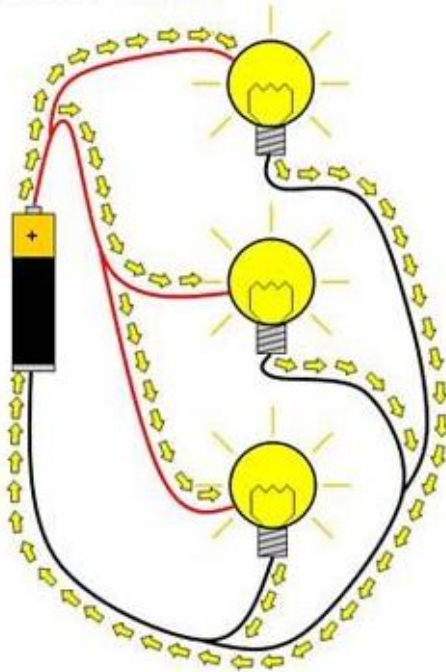
⚡ Example:

Two 2Ω resistors in series:

Total resistance = $2\Omega + 2\Omega = 4\Omega$

Parallel Circuits

Components are connected across the **same two nodes**, so the **same voltage** is applied to each.



🔑 Key Characteristics:

- **Voltage:** Same across all components

$$V_{\text{total}} = V_1 = V_2 = \dots$$

- **Current:** Splits between paths

$$I_{\text{total}} = I_1 + I_2 + \dots$$

- **Resistance:** Adds inversely

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

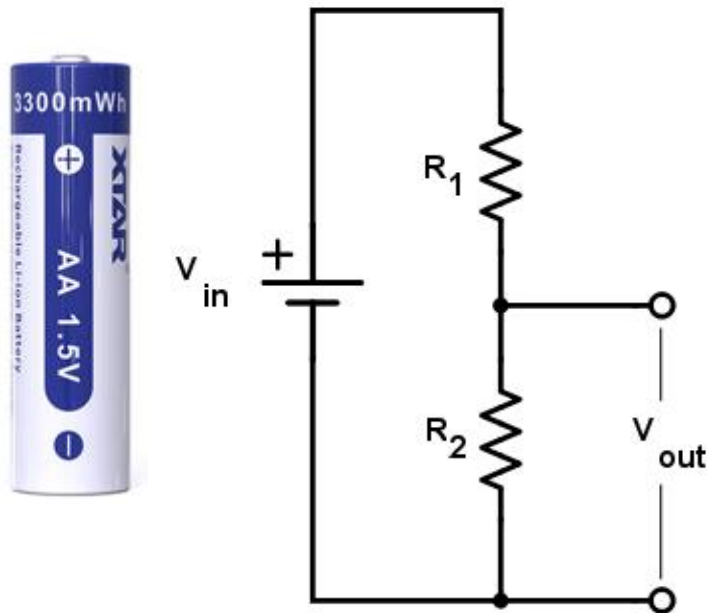
⚡ Example:

Two 2Ω resistors in parallel:

$$\frac{1}{R_{\text{total}}} = \frac{1}{2} + \frac{1}{2} = 1 \Rightarrow R_{\text{total}} = 1\Omega$$

Voltage Divider

A **series circuit** of resistors used to get a **reduced voltage** from a larger source.



Formula:

$$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$$

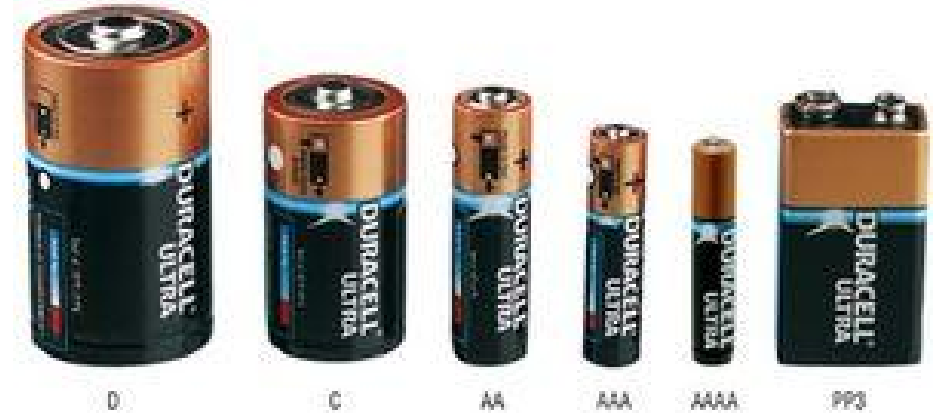
(Where V_{out} is across R_2)



Example:

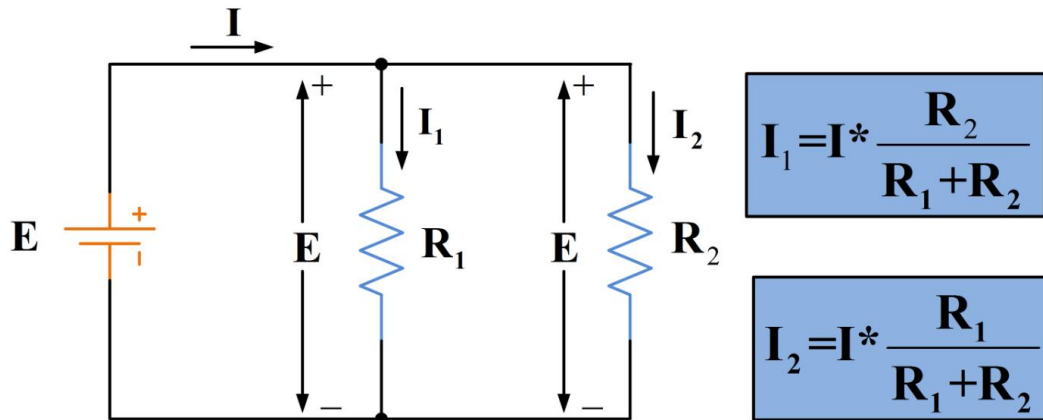
With 10V input, $R_1 = 2k\Omega$, $R_2 = 3k\Omega$:

$$V_{out} = 10 \times \frac{3000}{2000 + 3000} = 6V$$



Current Divider

A **parallel circuit** used to split current between branches.



Formula (for two resistors):

$$I_1 = I_{\text{total}} \times \frac{R_2}{R_1 + R_2}$$

(Current through one resistor depends on the resistance of the *other*!)



Example:

If $I_{\text{total}} = 4A$, $R_1 = 2\Omega$, $R_2 = 6\Omega$:

$$I_1 = 4 \times \frac{6}{2 + 6} = 3A$$

Exercise

A circuit is connected to a 24 V DC supply. It consists of:

$R1 = 4\ \Omega$ connected in series with a parallel combination of:

◦ $R2 = 6\ \Omega$

◦ $R3 = 3\ \Omega$

Questions:

1. Find the equivalent resistance of the parallel combination ($R2$ and $R3$).
2. Find the total resistance of the circuit.
3. Calculate the total current supplied by the battery.
4. Find the voltage across the parallel branch (use voltage divider concept).
5. Find the current through $R2$ and $R3$ (use current divider concept).
6. Verify that the sum of branch currents equals the total current.

Summary

Type	Voltage	Current	Resistance
Series	Divides	Same	Adds up
Parallel	Same	Divides	Adds inversely
Voltage Divider	Drops voltage	—	Based on resistor ratio
Current Divider	—	Splits current	Based on inverse resistance